

Calibration Certificate



Certificate No.: C131563 / L1030877

Calibration Object: Laser Vibrometer
Manufacturer: Polytec, GmbH
Controller/Front-End Type: VFX-F-110
Controller/Front End SN: 0331036
Sensor Head Type: VFX-I-160
Sensor Head SN: 0331024
Calibration Procedure: Vib_Calibration_1217_02
Date of Calibration: July 21, 2025
Calibration Due Date: July 21, 2027
Confirmed by customer? Yes
Technician: EMh
Calibration Location: Polytec, Inc
Customer: **Massachusetts Institute of Technology**
350 Brookline St
Middlesex
Cambridge, MA 02139
United States
 $\lambda=1550\text{ nm}$

Polytec, Inc
1 Cabot Road, Suite 102
Hudson, MA 01749
United States
Telephone: 508-417-1040
Fax: 508-281-4725

Traceability:

The calibrations within the certificate/report are traceable through NIST or another National Metrology Institute to the International System of Units (SI).

This calibration certificate includes the following protocols:

Page	Object	Procedure
2	VFX-V (X)	Part 1: Electrical Calibration
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4	VFX-D (X)	Part 1: Electrical Calibration
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6	VFX-I-160	Alignment of Head Protocol

Note: The statement of compliance in this certificate was issued without taking the uncertainty of measurement into consideration. The customer shall assess the results and uncertainty when determining if the results meet their needs.

Sign:

Elizabeth Mhawalla

EMh

Velocity Decoder Calibration Protocol



Part 1: Electrical Calibration

Certificate No.: C131563 / L1030877

Device Type: VFX-F-110

Serial No.: 0331036

Decoder Type: VFX-V (X)

Reference Conditions:

Frequency ¹ :	1 kHz	Measurement Procedure: Synthetic Doppler signals substitute those which would come from the vibrometer optical system when acquiring a sinusoidal vibration of known frequency and amplitude. The output of the respective decoder is compared to the represented reference vibration amplitude.
Amplitude ¹ :	70%	
Target Temp.:	(25 ± 3) °C	
Actual Temp.:	23.1 °C	
Uncertainty ² :	1.00%	

¹In some measurement ranges, deviating values are possible (refer to table)

²The reported expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor k=2, corresponding to a confidence level of approximately 95%.

Reference Instruments

Type:	Tektronix AFG 3252	Type:	Keithley 2001
Serial Number:	C022189	Serial Number:	0661057
Certificate No.:	3292055	Certificate No.:	3287284
Due Date:	Dec. 31, 2025	Due Date:	Dec. 31, 2025

Measurement Values

Range	Amplitude @ Frequency	Set Value	Actual Value
10.0 mm/s	7.0000 mm/s @ 1 kHz	0.9900 Vrms	0.9886 Vrms
20.0 mm/s	14.000 mm/s @ 1 kHz	0.9900 Vrms	0.9886 Vrms
50.0 mm/s	35.000 mm/s @ 1 kHz	0.9900 Vrms	0.9889 Vrms
100 mm/s	70.000 mm/s @ 1 kHz	0.9900 Vrms	0.9889 Vrms
200 mm/s	140.00 mm/s @ 1 kHz	0.9900 Vrms	0.9890 Vrms
500 mm/s	350.00 mm/s @ 1 kHz	0.9900 Vrms	0.9889 Vrms
1.00 m/s	700.00 mm/s @ 1 kHz	0.9900 Vrms	0.9890 Vrms
2.00 m/s	1.4000 m/s @ 1 kHz	0.9900 Vrms	0.9890 Vrms
5.00 m/s	3.5000 m/s @ 1 kHz	0.9900 Vrms	0.9889 Vrms
6.00 m/s	4.2000 m/s @ 1 kHz	0.9900 Vrms	0.9890 Vrms

Comments

Result of this Calibration Step:

N/A

Date
Jul. 21, 2025

Technician
EMh

Velocity Decoder Calibration Protocol



Part 2: Mechanical End Test

Certificate No.: C131563 / L1030877

Device Type: VFX-F-110

Serial No.: 0331036

Decoder Type: VFX-V (X)

Reference Conditions:

Frequency¹:	159.17 Hz	Measurement Procedure: The laser beam of the vibrometer is directed to the reference surface of a vibration calibrator, which is traceable to national standards for the realization of the physical units. The output of the respective decoder is compared to the reference vibration amplitude.
Amplitude¹:	10.24 mm/s	
Target Temp.:	(25 ± 3) °C	
Actual Temp:	23.1 °C	
Uncertainty²:	3.00%	

¹In some measurement ranges, deviating values are possible (refer to table)

²The reported expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor k=2, corresponding to a confidence level of approximately 95%.

Reference Instruments

Type:	Keithley 2001	Type:	Metra VC-20
Serial Number:	0661057	Serial Number:	220550
Certificate No.:	3287284	Certificate No.:	52209
Due Date:	Dec. 31, 2025	Due Date:	Dec. 31, 2025

Measurement Values

Range	Amplitude @ Frequency	Set Value	Actual Value
20.0 mm/s	10.24 mm/s @ 159.17 Hz	1.0240 Vrms	1.0020 Vrms
50.0 mm/s	10.24 mm/s @ 159.17 Hz	0.4096 Vrms	0.4003 Vrms
100 mm/s	10.24 mm/s @ 159.17 Hz	0.2048 Vrms	0.2005 Vrms
200 mm/s	10.24 mm/s @ 159.17 Hz	0.1024 Vrms	0.1002 Vrms
500 mm/s	10.24 mm/s @ 159.17 Hz	0.0410 Vrms	0.0401 Vrms

Comments

Result of this Calibration Step:

No adjustments to the decoder were necessary

Date
Jul. 21, 2025

Technician
EMh

Displacement Decoder Calibration Protocol



Part 1: Electrical Calibration

Certificate No.: C131563 / L1030877

Device Type: VFX-F-110

Serial No.: 0331036

Decoder Type: VFX-D (X)

Reference Conditions:

Frequency ¹ :	1 kHz	Measurement Procedure: Synthetic Doppler signals substitute those which would come from the vibrometer optical system when acquiring a sinusoidal vibration of known frequency and amplitude. The output of the respective decoder is compared to the represented reference vibration amplitude.
Amplitude ¹ :	70%	
Target Temp.:	(25 ± 3) °C	
Actual Temp.:	23.0 °C	
Uncertainty ² :	1.00%	

¹In some measurement ranges, deviating values are possible (refer to table)

²The reported expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor k=2, corresponding to a confidence level of approximately 95%.

Reference Instruments

Type:	Tektronix AFG 3252	Type:	Keithley 2001
Serial Number:	C022189	Serial Number:	0661057
Certificate No.:	3292055	Certificate No.:	3287284
Due Date:	Dec. 31, 2025	Due Date:	Dec. 31, 2025

Measurement Values

Range	Amplitude @ Frequency	Set Value	Actual Value
500 nm	481.25 nm @ 1 kHz	1.3612 Vrms	1.3624 Vrms
1.00 µm	700.00 nm @ 1 kHz	0.9900 Vrms	0.9911 Vrms
2.00 µm	1.4000 µm @ 1 kHz	0.9900 Vrms	0.9908 Vrms
5.00 µm	3.5000 µm @ 1 kHz	0.9900 Vrms	0.9905 Vrms
10.0 µm	7.0000 µm @ 1 kHz	0.9900 Vrms	0.9905 Vrms
20.0 µm	14.000 µm @ 1 kHz	0.9900 Vrms	0.9907 Vrms
50.0 µm	35.000 µm @ 1 kHz	0.9900 Vrms	0.9906 Vrms
100 µm	70.000 µm @ 1 kHz	0.9900 Vrms	0.9906 Vrms
200 µm	140.00 µm @ 1 kHz	0.9900 Vrms	0.9906 Vrms
500 µm	350.00 µm @ 1 kHz	0.9900 Vrms	0.9907 Vrms
1.00 mm	700.00 µm @ 1 kHz	0.9900 Vrms	0.9907 Vrms
2.00 mm	1.4000 mm @ 500 Hz	0.9900 Vrms	0.9912 Vrms
5.00 mm	3.5000 mm @ 250 Hz	0.9900 Vrms	0.9913 Vrms
10.0 mm	7.0000 mm @ 125 Hz	0.9900 Vrms	0.9911 Vrms
20.0 mm	14.000 mm @ 62 Hz	0.9900 Vrms	0.9909 Vrms
50.0 mm	35.000 mm @ 15 Hz	0.9900 Vrms	0.9908 Vrms
100 mm	70.000 mm @ 10 Hz	0.9900 Vrms	0.9908 Vrms
200 mm	70.000 mm @ 10 Hz	0.4950 Vrms	0.4952 Vrms

Comments

Result of this Calibration Step:

No adjustments to the decoder were necessary

Date
Jul. 21, 2025

Technician
EMh

Displacement Decoder Calibration Protocol



Part 2: Mechanical End Test

Certificate No.: C131563 / L1030877

Device Type: VFX-F-110

Serial No.: 0331036

Decoder Type: VFX-D (X)

Reference Conditions:

Frequency¹:	159.17 Hz	Measurement Procedure: The laser beam of the vibrometer is directed to the reference surface of a vibration calibrator, which is traceable to national standards for the realization of the physical units. The output of the respective decoder is compared to the reference vibration amplitude.
Amplitude¹:	10.24 μ m	
Target Temp.:	(25 $\pm$ 3) °C	
Actual Temp:	22.3 °C	
Uncertainty²:	3.00%	

¹In some measurement ranges, deviating values are possible (refer to table)

²The reported expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor k=2, corresponding to a confidence level of approximately 95%.

Reference Instruments

Type:	Keithley 2001	Type:	Metra VC-20
Serial Number:	0661057	Serial Number:	220550
Certificate No.:	3287284	Certificate No.:	52209
Due Date:	Dec. 31, 2025	Due Date:	Dec. 31, 2025

Measurement Values

Range	Amplitude @ Frequency	Set Value	Actual Value
20.0 μ m	10.24 μ m @ 159.17 Hz	1.0240 Vrms	1.0028 Vrms
50.0 μ m	10.24 μ m @ 159.17 Hz	0.4096 Vrms	0.4010 Vrms
100 μ m	10.24 μ m @ 159.17 Hz	0.2048 Vrms	0.2004 Vrms
200 μ m	10.24 μ m @ 159.17 Hz	0.1024 Vrms	0.1003 Vrms
500 μ m	10.24 μ m @ 159.17 Hz	0.0410 Vrms	0.0400 Vrms

Comments

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Result of this Calibration Step:

No adjustments to the decoder were necessary

Date
Jul. 21, 2025

Technician
EMh

Alignment of Head Protocol



Certificate No.: C131563 / L1030877

Device Type: VFX-I-160

Serial No.: 0331024

Optics Check: A separate test is done to check interferometer performance. Various check points are tested to assure the interferometer meets manufacturer specifications. The optical sensitivity is optimized by aligning the interferometer and maximizing signal beam power. The output beam(s) are then checked on a test surface for maximum signal level and a clean, stable signal.

Reference Instruments

Type:	Thor Labs PM100A	Type:	Thor Labs 132C
Serial Number:	P1001255	Serial Number:	220920316
Certificate No.:	491451882025228	Certificate No.:	491451882025228
Due Date:	Feb. 28, 2026	Due Date:	Feb. 28, 2026

Laser Tube:	
1) Output power of beam	9.00 mW
2) Beam is centered in aperture of lens	Pass
3) Beam has correct continuous profile	Pass
Optical Alignment:	
4) Pilot laser and IR laser superimposed at 3+ meters	Pass
5) Pilot laser and IR laser superimposed at short range	Pass
6) Alignment of detector modules	Pass
End Test:	
7) Background signal level without surface reflections	Pass
8) Velocity signal is free of noise and instabilities	Pass

- ☒ PASSED VERIFIED
☐ FAILED VERIFIED

Comments

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Date
Jul. 21, 2025

Technician
EMh